

# Sankaran Vaidyanathan

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## Research Interests

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Developing causal inference methods, particularly at the instance-specific and path-specific level, for mechanistic interpretability and evaluation of complex AI systems such as LLMs and RL agents.

## Education

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| <b>University of Massachusetts Amherst</b><br><i>Ph.D. in Computer Science</i>  | <i>Sept 2021–May 2027</i> |
| <b>University of Massachusetts Amherst</b><br><i>M.S. in Computer Science</i>   | <i>Sept 2019–May 2024</i> |
| <b>Anna University</b><br><i>B.E. in Electrical and Electronics Engineering</i> | <i>Aug 2013–May 2017</i>  |

## Experience

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| <b>Research Intern</b><br><i>Basis Research Institute</i>                                                                                                            | <i>Cambridge, MA</i><br><i>June 2026–present</i>    |
| <b>Research Fellow</b><br><i>AI Safety Camp</i>                                                                                                                      | <i>Remote</i><br><i>Feb 2025–Aug 2025</i>           |
| <b>Research Assistant</b><br><i>Knowledge Discovery Lab &amp; Safe, Confident, and Aligned Learning + Robotics Lab</i><br><i>University of Massachusetts Amherst</i> | <i>Amherst, MA</i><br><i>May 2020–present</i>       |
| <b>Project Associate</b><br><i>Robert Bosch Center for Data Science and AI</i><br><i>Indian Institute of Technology Madras</i>                                       | <i>Chennai, India</i><br><i>July 2017–June 2019</i> |

## Selected Publications

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### **The Curse of Multiple Mediators: Hidden Interaction Effects in Activation Patching**

**Sankaran Vaidyanathan**, David Arbour, Aaron Mueller, Scott Niekum, David Jensen  
*arXiv:2606.27510 (under review)*

### **Hierarchical Experimentalist Agents**

Abhranil Chandra, **Sankaran Vaidyanathan**, Utsav Dhanuka, Varun Gandhi, Scott Niekum  
*RLC Workshop on Reinforcement Learning Beyond Rewards: Towards Scalable General-Purpose Agents, 2026*

### **Detecting and Characterizing Planning in Language Models**

Jatin Nainani, **Sankaran Vaidyanathan**, Connor Watts, Andre N. Assis, Alice Rigg  
*NeurIPS Workshop on Mechanistic Interpretability, 2025*

### **Judging the Judges: Evaluating Alignment and Vulnerabilities in LLMs-as-Judges**

Aman Singh Thakur\*, Kartik Choudhary\*, Venkat Srinik Ramayapally\*, **Sankaran Vaidyanathan**, Dieuwke Hupkes  
*ACL Workshop on Natural Language Generation, Evaluation, and Metrics (GEM), 2025*

### **Quantitative LLM Judges**

Aishwarya Sahoo\*, Jeevana Kruthi Karnuthala\*, Tushar Parmanand Budhwani\*, Pranchal Agarwal\*, **Sankaran Vaidyanathan**, Alexa Siu, Franck Dernoncourt, Jennifer Healey, Nedim Lipka, Ryan Rossi, Uttaran Bhattacharya, Branislav Kveton  
*arXiv:2506.02945 (under review)*

### **Automated Discovery of Functional Actual Causes in Complex Environments**

Caleb Chuck\*, **Sankaran Vaidyanathan**\*, Stephen Giguere, Amy Zhang, David Jensen, Scott Niekum  
*arXiv:2404.10883 (under review)*

## Selected Projects

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### Probabilistic Actual Causality in Transformers

June 2026–present

Basis Research Institute

- Identifying redundant mechanisms in LLMs using actual causality and probabilistic programming.

### Interphyre

Sep 2025–May 2026

[github.com/sankaranv/interphyre](https://github.com/sankaranv/interphyre)

- Built a 2D physics puzzle environment for interpretability and agentic experimentation research, with levels as Python code, full simulation state access, and an intervention API for paired counterfactual trajectories.

### Evaluation Awareness in LLM Safety Benchmarks

Sep 2025–May 2026

Microsoft

- Showed that LLMs internally represent whether or not they are being tested even when they fail to verbalize it, and that models better at detecting tests are harder to manipulate into producing harmful outputs.

### Quantitative LLM Judges

Feb 2025–May 2025

Adobe Research

- Developed a lightweight post-hoc framework that improves LLM-as-a-Judge alignment with human scores using generalized linear models on judge embeddings, avoiding expensive fine-tuning.

### Evaluating Alignment and Vulnerabilities in LLMs-as-Judges

Feb 2024–Dec 2024

Meta

- Demonstrated that simple lexical metrics and smaller models can match large LLM judges at ranking model outputs, and that standard agreement metrics mask systematic misalignment with human scores.

### Analysis and Prediction of Cognitive Load During Cardiac Surgery

May 2023–May 2024

National Institute of Health and Harvard Medical School

- Predicted cognitive load in surgical teams from time-series heart rate variability data, and identified key predictive features using SHAP and feature ablation validated against clinical expertise.

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## Teaching Experience

- Guest Lecture: Mechanistic Interpretability, *COMPSCI 690S: AI Alignment*

Spring 2026

- Guest Lecture: MCMC and Hamiltonian Monte Carlo, *MATH 605: Probability Theory*

Fall 2022

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## Service and Outreach

- Reviewer: NeurIPS 2026, ICML 2026, AAAI 2026, NeurIPS 2025 Mechanistic Interpretability Workshop

- Co-organizer, UMass Machine Learning and Friends Lunch

2019–2020, 2023–2026

- Founded UMass Mechanistic Interpretability Reading Group

2025–2026

- Mentor, UMass Ph.D. Applicant Support Program

2021–2025

- M.S. Graduate Representative, UMass College of Information and Computer Sciences

2020

- Volunteer Pen-Pal, Letters to a Pre-Scientist

2024–2026

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## Technical Skills

**Languages and Frameworks:** Python, C++, PyTorch, Pyro-PPL, ChiRho, Box2D, NNsight

**Design:** Figma, Inkscape, Affinity Designer, Adobe After Effects

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## Relevant Graduate Coursework

Bayesian Statistics, Causal Inference, Probabilistic Graphical Models, Reinforcement Learning, Machine Learning, Artificial Intelligence, Advanced Natural Language Processing, Neural Networks: A Modern Introduction, Optimization in Computer Science, Math Statistics, Research Methods in Empirical Computer Science, Probability Theory, Distributed and Operating Systems, Quantum Information Systems, Fixing Social Media